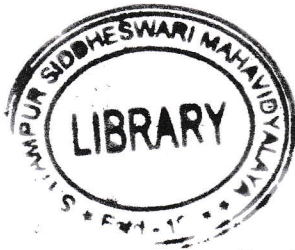


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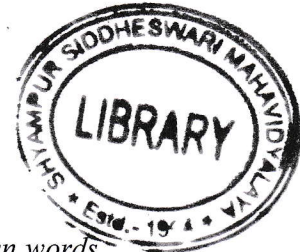
COMPUTER SCIENCE — GENERAL



Paper : SEC-A-X-2

(Software Engineering)

Full Marks : 80



The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question nos. 1 and 2** and **any four** from the rest.

1. Answer **any ten** questions :

2×10

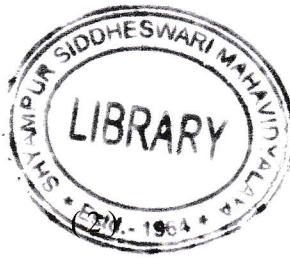
- (a) Why is feasibility study performed?
- (b) Write the advantages of using Prototype.
- (c) Why is β -testing performed?
- (d) What is context diagram?
- (e) What is cohesion?
- (f) How many level-0 and level-1 DFD may exists in a system?
- (g) What do you understand by open systems?
- (h) What do you understand by software fault?
- (i) What is unit testing?
- (j) Why is risk analysis important?
- (k) Why is maintenance required?
- (l) What are the drawbacks of waterfall model?
- (m) What are the different phases of SDLC?
- (n) Define : Software Engineering.
- (o) Why is requirement analyse important in software development?

2. Write short notes on **any four** :

5×4

- (a) Software Quality Assurance
- (b) Decision Tree
- (c) System Testing
- (d) SRS
- (e) COCOMO.

Please Turn Over



X(3rd Sm.)-Computer Sc.-G/SEC-A-X-2/CBCS

3. (a) Explain iterative waterfall model in detail.
(b) List any four prime objectives of software engineering. 6+4
 4. (a) What are the advantages and disadvantages of spiral model?
(b) Describe the functionality of a spiral round. (3+2)+5
 5. (a) Differentiate between cohesion and coupling.
(b) Explain physical DFD with example.
(c) What are the different types of software errors? 3+5+2
 6. (a) Draw a data flow diagram for a library management system.
(b) Distinguish between software verification and software validation. 6+4
 7. (a) What is data dictionary? How is it used in software engineering?
(b) Compare evolutionary and throw away prototyping. (2+4)+4
 8. (a) Why is testing required?
(b) What is the significance of Integration testing?
(c) Explain Black box testing with example. 2+2+6
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